

Introduction to Machine Learning

Mamta Bhattarai

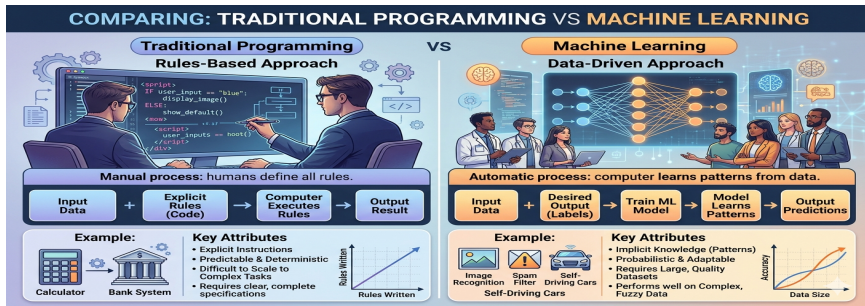
MBUST

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What is Machine Learning?

- Machine Learning (ML) is a branch of Artificial Intelligence (AI).
- It enables computers to learn patterns from data without being explicitly programmed.
- ML systems improve their performance automatically through experience.
- Introduced by Arthur Samuel (1959):
“Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed.”

Traditional Programming vs. Machine Learning



Traditional Programming

Rules are explicitly programmed

Input + Rules → Output

Works well for structured tasks

Deterministic behavior

Calculator, Banking System

Machine Learning

Rules are learned from data

Input + Output → Model

Handles complex and unstructured tasks

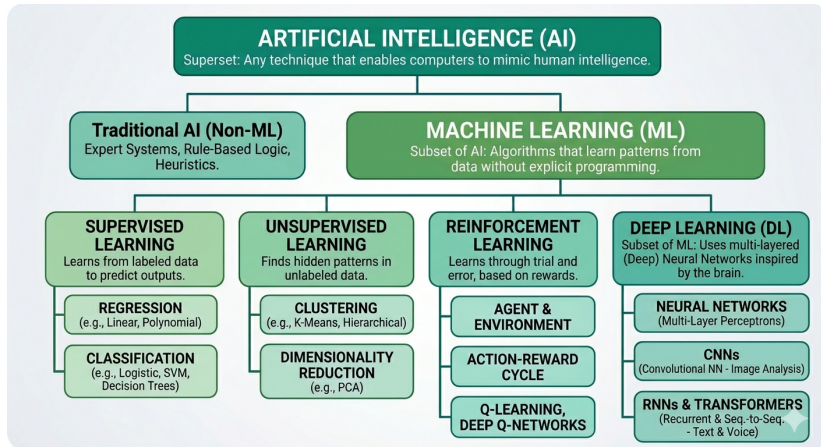
Probabilistic predictions

Face Recognition, Chatbots

History of Machine Learning

- Machine Learning (ML) originated in the 1950s as a subfield of Artificial Intelligence (AI).
- In 1952, **Arthur Samuel** introduced a self-learning checkers program and coined the term “Machine Learning”.
- In 1957, **Frank Rosenblatt** proposed the Perceptron, an early neural network model.
- 1970s–1980s: Development of rule-based expert systems.
- 1990s: Emergence of statistical learning methods such as decision trees and support vector machines.
- 2000s–present: Growth of big data and computing power enabled deep learning and advanced AI applications.
- Today, ML is widely used in healthcare, finance, robotics, and many other domains.

Artificial Intelligence and Machine Learning



- **Artificial Intelligence (AI):** Broad field of creating intelligent systems.
- **Machine Learning (ML):** Subset of AI focused on learning from data.

Relationship Among AI, ML and Data Mining

Artificial Intelligence

Develops systems capable of intelligent decision-making.

Machine Learning

Provides algorithms that learn from data.

Data Mining

Discovers hidden patterns, trends and relationships in data.

Why Machine Learning?

- Handles large volumes of data efficiently.
- Identifies hidden patterns and relationships.
- Automates decision-making processes.
- Improves prediction accuracy over time.
- Enables personalized user experiences.

Common Applications of Machine Learning

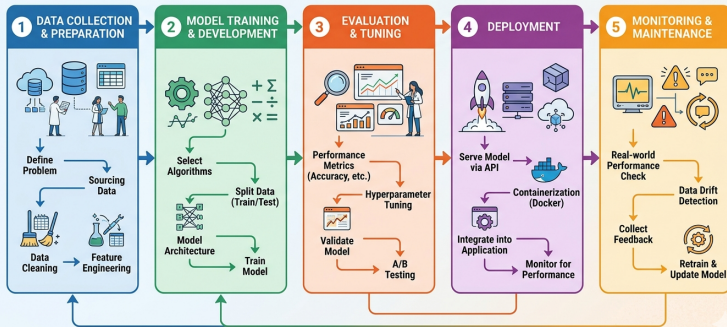
- Image Recognition
- Speech Recognition
- Natural Language Processing
- Recommendation Systems
- Fraud Detection
- Medical Diagnosis
- Autonomous Vehicles
- Predictive Analytics

Applications in Different Sectors

Sector	Applications
Healthcare	Disease diagnosis, medical imaging
Finance	Fraud detection, risk assessment
Education	Personalized learning
Agriculture	Crop monitoring, yield prediction
Retail	Product recommendation
Transportation	Autonomous vehicles, route planning
Cybersecurity	Intrusion detection
Manufacturing	Predictive maintenance

Machine Learning Pipeline

UNDERSTANDING THE MACHINE LEARNING WORKFLOW



Data → Preprocessing → Feature Engineering → Training → Evaluation → Deployment

- **Data:** Collect raw data from databases, sensors, web applications, surveys, or other sources.
- **Preprocessing:** Clean the data by handling missing values, removing duplicates, normalizing features, and encoding categorical variables.
- **Feature Engineering:** Create informative features from raw data using domain knowledge. Examples include extracting date information, generating interaction features, or computing ratios such as BMI. Well-designed features often improve performance more than using a more complex algorithm.
- **Training:** Train a machine learning model on the prepared dataset to learn patterns and relationships.
- **Evaluation:** Assess model performance using metrics such as Accuracy, Precision, Recall, F1-score, RMSE, or AUC.
- **Deployment:** Integrate the trained model into real-world applications where it can make predictions on new data.

Types of Machine Learning

Type	Goal
Supervised Learning	Learn from labeled data
Unsupervised Learning	Discover patterns in unlabeled data
Semi-Supervised Learning	Use both labeled and unlabeled data
Reinforcement Learning	Learn through rewards and penalties

Example Dataset for Different Learning Types

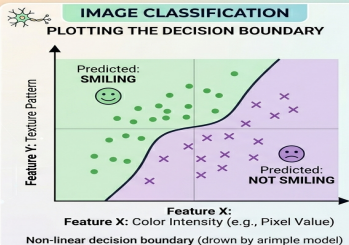
Student	Study Hours	Attendance (%)	Final Grade
S1	8	95	A
S2	6	85	B
S3	2	60	C
S4	5	80	?
S5	7	90	?

How Different Learning Types Use This Dataset

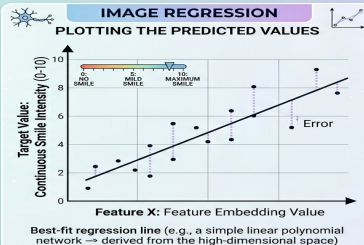
- **Supervised Learning:** Use labeled students (S1–S3) to predict the grades of S4 and S5.
- **Unsupervised Learning:** Ignore grades and group students into clusters based on study hours and attendance.
- **Semi-Supervised Learning:** Use labeled data (S1–S3) together with unlabeled data (S4–S5) to improve prediction accuracy.
- **Reinforcement Learning:** An intelligent tutoring system recommends study plans and receives rewards when student performance improves.

Supervised Learning

SIMPLE PLOTTING EXAMPLES: VISUALIZING THE OUTPUTS



Simple neuron, dataset



Best-fit regression line

- Training data contains input–output pairs.
- Learns a mapping from inputs to outputs.
- Main tasks:
 - Classification: Predicts discrete class labels.
 - Regression: Predicts continuous numerical values.

Examples of Supervised Learning

Classification

Predict discrete labels.

- Spam vs Non-spam Email
- Disease Detection

Regression

Predict continuous values.

- House Price Prediction
- Stock Price Forecasting

Classification vs Regression

Classification

- Predict categorical labels
- Output is discrete
- Example: Spam/Not Spam

Regression

- Predict numerical values
- Output is continuous
- Example: House Price

Unsupervised Learning

- Works with unlabeled data, where no target labels are provided.
- Discovers hidden structures, patterns, and relationships in data.
- Common tasks:
 - Clustering
 - Dimensionality Reduction
 - Association Rule Mining
- Applications:
 - Customer Segmentation
 - Market Basket Analysis
 - Document Grouping
 - Anomaly Detection

Key Idea: Unsupervised learning identifies meaningful patterns without using labeled examples.

Illustration of Unsupervised Learning

UNSUPERVISED LEARNING METHODS

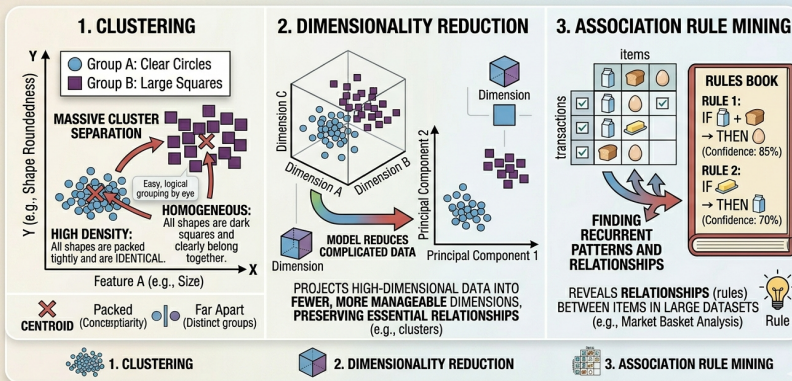


Figure: In unsupervised learning, similar data points are automatically grouped into clusters based on their characteristics without prior class labels.

Supervised vs Unsupervised Learning

Supervised Learning	Unsupervised Learning
Uses labeled data	Uses unlabeled data
Predicts outcomes	Finds hidden patterns
Classification and Regression	Clustering and Association
Higher prediction accuracy	Useful for exploratory analysis

Advantages and Challenges

Advantages

- Automation of complex tasks
- Better decision making
- Improved accuracy
- Scalability

Challenges

- Data quality issues
- Overfitting
- Computational cost
- Model interpretability

Summary

- Machine Learning enables systems to learn from data.
- ML is a subset of Artificial Intelligence.
- Data mining focuses on extracting knowledge from data.
- ML has applications across healthcare, finance, education, agriculture, and more.
- Major learning paradigms include supervised and unsupervised learning.
- A typical ML workflow includes data collection, preprocessing, training, evaluation, and deployment.

Thank You